

Socio-Economic Factors Influencing Cancer Mortality in the United States: An Investigation using Multiple Linear Regression Analysis

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Abstract—Cancer is a major cause of mortality in the United States, and there are many socio-economic factors that may contribute to this issue. In this paper, I investigate the relationship between socio-economic factors and cancer mortality using multiple linear regression analysis. My study is based on data from provided to us on Moodle, which includes information on cancer incidence, deathrate, and socio economic characteristics of the U.S. population.

I found that factors such as income, education, and place of origin are significant predictors of cancer death rates, after controlling for other related variables. My results suggest that addressing socio-economic disparities in cancer prevention and treatment may be an important public health priority. [1]

Index Terms—Multiple Regression, Cancer, death rate predictions

I. INTRODUCTION

In In this report, I look at building a multiple linear regression model using given cancer dataset. This dataset was provided by NCI as a CSV file named cancer.csv.

Multiple linear regression is a statistical technique used to model the relationship between multiple predictor variables and a single response variable. The goal of this technique is to create a linear equation that predicts the value of the response variable based on the values of the predictor variables [3].

In this project, I want to use the socio-economic variables to predict the cancer death rate. The dataset contains 3047 observations with 25 variables.

The first variable is `County`, a character vector that identifies the county in which each observation was taken.

The second variable is `Population`, an integer vector indicating the population of each county.

The third variable is `deathRate`, a numeric vector that indicates the death rate per 100,000 people in each county.

The fourth variable is `incidenceRate`, a numeric vector that indicates number of newly diagnosed cancer cases per 100,000 people in each county.

The remaining variables are related to socio-economic factors, such as median income, poverty rate, education level, age, and race.

The general equation for multiple linear regression is:

Where:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Fig. 1. General equation for multiple linear regression [2].

Y is the response variable (in this case, the cancer death rate)

β_0 is the intercept

$\beta_1, \beta_2, \dots, \beta_p$ are the coefficients of the independent variables ($X_1, X_2, \dots, X_p$)

ϵ is the error term.

The goal of multiple linear regression is to estimate the values of $\beta_0, \beta_1, \beta_2, \dots, \beta_p$ that minimize the sum of squared errors between the predicted values and the actual values of the response variable [3].

II. DATA PRE-PROCESSING

A. Structure of DataSet

The dataset contains information about various demographic and socio-economic related variables for 3,047 counties in the United States. The variables include population, death rate, incidence rate, median income, poverty percent, median age, household size, percentage of married households, education level, unemployment rate, and percentage of different racial groups.

Table I displays the descriptive summary of the dataset. I can see that some variables have outliers. `MedianAge` variable have maximum value as 600+ i.e. not possible in real life.

TABLE I
SUMMARY OF DATASET

Variable	min	median	mean	max
Population	827	26643	102637	10170292
deathRate	59.7	178.1	178.7	362.8
incidenceRate	201.3	449.5	445.7	1206.9
medIncome	22640	45207	47063	125635
povertyPercent	3.20	15.90	16.88	47.40
MedianAge	22.30	41.00	45.27	624.00
MedianAgeMale	22.40	39.60	39.57	64.70
MedianAgeFemale	22.30	42.40	42.15	65.70
AvgHouseholdSize	1.86	2.50	2.53	3.97
PctMarriedHouseholds	22.99	51.67	51.24	78.08
PctNoHS18_24	0.00	17.10	18.22	64.10
PctHS18_24	0.0	34.7	35.0	72.5
PctBachDeg18_24	0.000	5.400	6.158	51.800
PctHS25_Over	7.50	35.30	34.80	54.80
PctBachDeg25_Over	2.50	12.30	13.28	42.20
PctUnemployed16_Over	0.400	7.600	7.852	29.400
PctPrivateCoverage	22.30	65.10	64.35	92.30
PctEmpPrivCoverage	13.5	41.1	41.2	70.7
PctPublicCoverage	11.20	36.30	36.25	65.10
PctPublicCoverageAlone	2.60	18.80	19.24	46.60
PctWhite	10.20	90.06	83.65	100.00
PctBlack	0	2.2476	9.1080	85.9478
PctAsian	0	0.5498	1.2540	42.6194
PctOtherRace	0	0.8262	1.9835	41.9303

B. Missing Data

I checked for NAs and NAN in the dataset. There was no missing data found in the given dataset.

III. DISTRIBUTION OF DATA

The distribution of data refers to the way that the values in a dataset are spread out or arranged. Understanding the distribution of a dataset is important in statistical analysis because it can help to identify patterns, outliers, and potential issues with the data [4].

A. Using Histogram

A histogram is a graphical representation of the distribution of a dataset that uses bars to show the frequency of occurrence of values within certain ranges. It provides a visual representation of the shape, center, and spread of the data [4].

A density curve, on the other hand, is a smooth curve that represents the distribution of a dataset. It shows the probability of values occurring at each point along the x-axis. Density curves are often used to compare the shapes of different distributions or to identify patterns and trends in the data [4].

In this section, I checked that if all my features data follows normal distribution or not.

Figure 2 to 5 displays the histogram with density Curve for all variables except County.

I found out that all data are having normal distribution except PctAsian, PctBlack, PctWhite in Fig. 3 and PctOtherRace, population, MedianAge in Fig. 2.

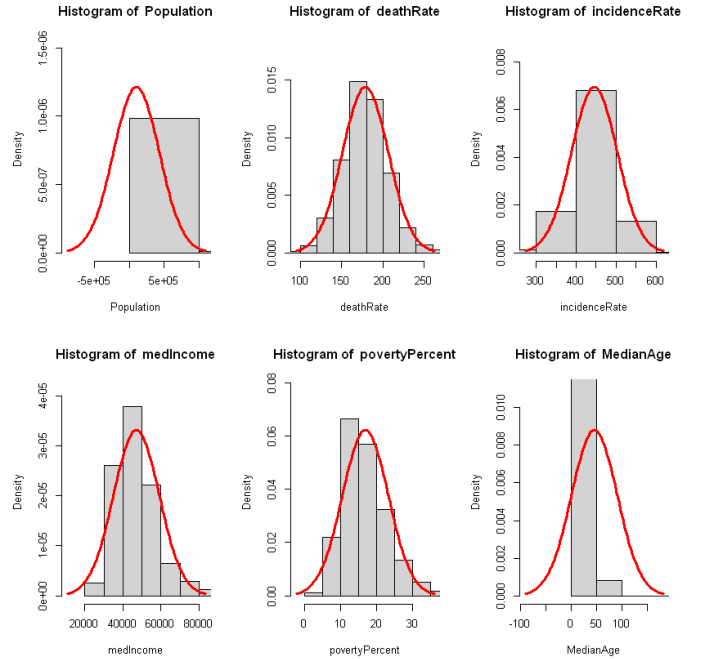


Fig. 2. Histogram with Density curve 1

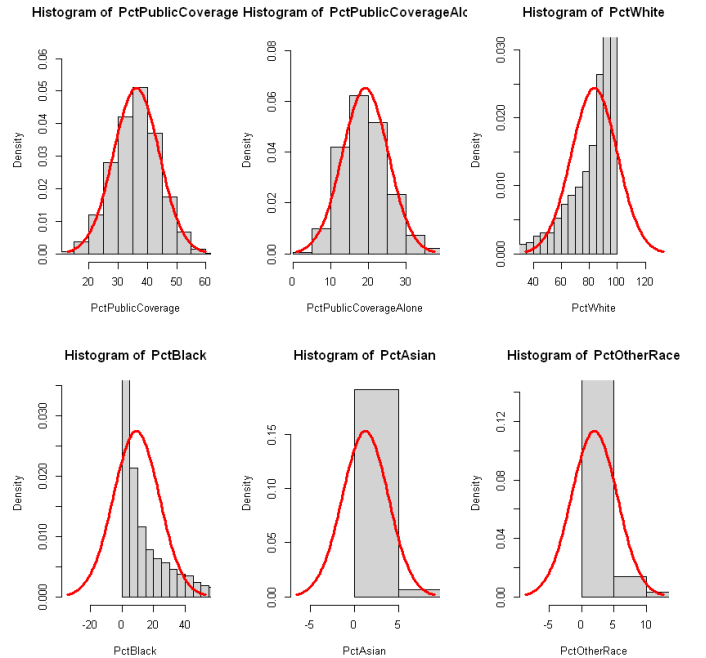


Fig. 3. Histogram with Density curve 2

B. Using Scatter plot

Using Scatter plot, I am checking the Correlation between the data or any outliers present or not. I found out that MedianAge, MedianAgeFemale and MedianAgeMale have some outliers and errors in data. As displayed in Fig.4,

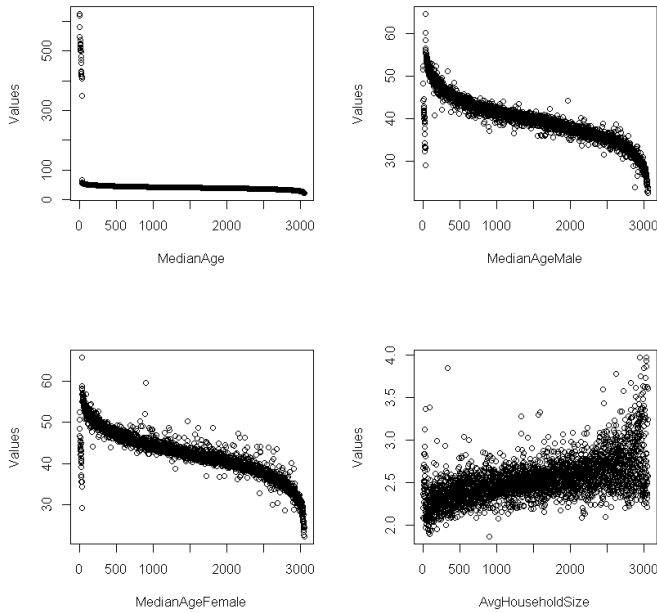


Fig. 4. Scatter plot of MedianAge, MedianAgeFemale and MedianAgeMale

IV. TRANSFORMATION OF DATA

On the basis of outliers and errors i found in above section. I am trying to transform the data so that it makes more sense during model creation.

A. PctBachDeg18_24

I found out that PctBachDeg18_24 data is right skewed. Hence i am transforming the data so that it follow normal distribution.

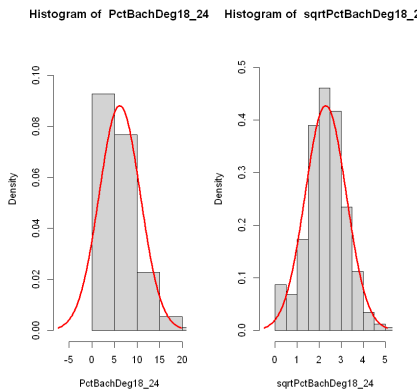


Fig. 5. Transformation of PctBachDeg18_24

Fig. 5 shows before and after transformation of the data.

I have used `sqrt()` function to transform the data and then saved it into `sqrtPctBachDeg18_24` column.

I then removed the `PctBachDeg18_24` column from the data frame to remove correlation with transformed data.

B. Population

I found out that Population data is not spread across the curve. Hence i am transforming the data so that it follow normal distribution.

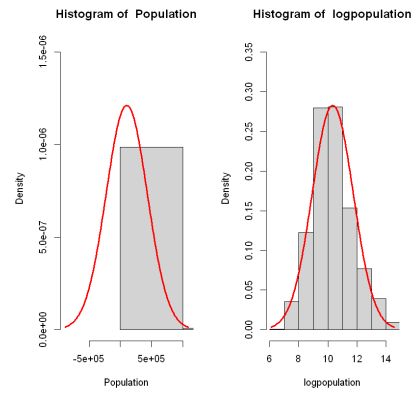


Fig. 6. Transformation of Population

Fig. 6 shows before and after transformation of the data.

I have used `log()` function to transform the data and then saved it into `logpopulation` column.

I then removed the `Population` column from the data frame to remove correlation with transformed data.

C. PctWhite

- **Before Transformation:** left skewed data.
- **Technique Used:** $PctWhite^{8.3}$.
- **New Column after transformation:** `PowPctWhite`.
- **Column removed after transformation:** `PctWhite`.

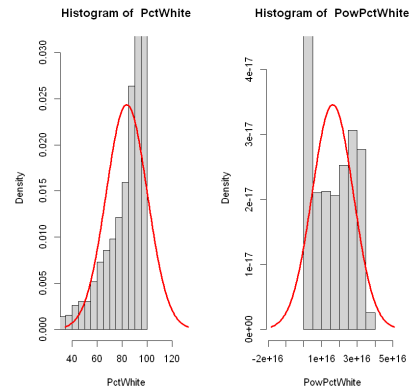


Fig. 7. Transformation of PctWhite

Fig. 7 shows before and after transformation of the data.

D. PctAsian

- **Before Transformation:** left skewed data.
- **Technique Used:** $\text{PctAsian}^{0.3}$.
- **New Column after transformation:** PowPctAsian.
- **Column removed after transformation:** PctAsian.

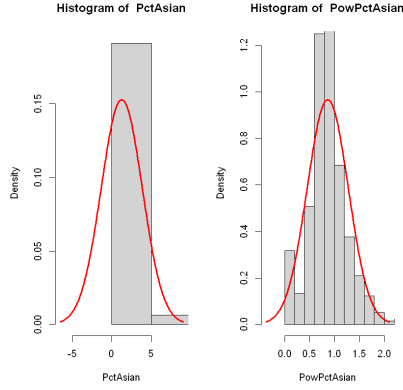


Fig. 8. Transformation of PctAsian

Fig. 8 shows before and after transformation of the data.

E. PctBlack

- **Before Transformation:** Right skewed data.
- **Technique Used:** $\text{PctBlack}^{0.20}$.
- **New Column after transformation:** PowPctBlack.
- **Column removed after transformation:** PctBlack.

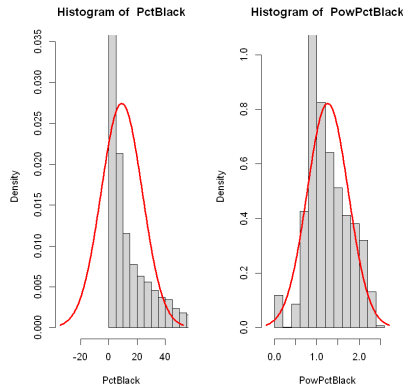


Fig. 9. Transformation of PctBlack

Fig. 9 shows before and after transformation of the data.

F. PctOtherRace

- **Before Transformation:** Right skewed data.
- **Technique Used:** $\text{PctOtherRace}^{0.24}$.
- **New Column after transformation:** PowPctOtherRace.
- **Column removed after transformation:** PctOtherRace.

Fig. 10 shows before and after transformation of the data.

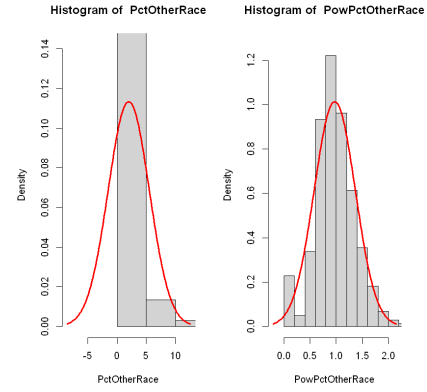


Fig. 10. Transformation of PctOtherRace

G. MedianAge, MedianAgeMale, MedianAgeFemale

For MedianAge, in fig. 4, Some Age values is greater than 100 so i have done below steps for transformation :

- Calculated mean of MedianAge where MedianAge < 100
- Replacing mean where MedianAge > 300

For MedianAgeMale and MedianAgeFemale, in fig. 4, Some MedianAgeMale, MedianAgeFemale values is less than or equal to 0 so i have done below steps for transformation :

- calculated mean of MedianAgeMale, MedianAgeFemale where MedianAgeMale or MedianAgeFemale > 0
- Replaced mean where MedianAgeMale or MedianAgeFemale ≤ 0

V. FEATURE SELECTION

Multicollinearity refers to a statistical occurrence where two or more predictor variables within a regression model exhibit a high level of correlation. As a result, the predictor variables fail to provide distinct or autonomous insights to the model, since they essentially capture the same underlying concept or trait.

in this section, I am selecting features for linear model.

A. Using Correlation function

correlation matrix alone does not provide information about the statistical significance of each relationship. Correlation coefficients may be influenced by outliers, non-linear relationships, and other factors that are not captured by the correlation matrix alone.

We can See outliers are Present in scatter plot above fig. 4. we cannot rely on Correlation for feature selection.

I have considered features having absolute value of **correlation greater than 0.4**.

Based on the correlation coefficients with the deathRate variable, the following features have relatively strong correlations:

incidenceRate (0.48) medIncome (-0.47)
povertyPercent (0.45) PctHS25_Over (0.42)

```
PctBachDeg25_Over (-0.5)
PctUnemployed16_Over (0.41)
PctPrivateCoverage (-0.41)
PctPublicCoverage (0.40)
PctPublicCoverageAlone (0.45)
```

These features are good candidates to include in our regression model.

Output of model `lm_Cor_Based` with above features suggests below : The variables with the most significant coefficients are incidence rate (which has a positive coefficient), poverty percent, `PctHS25_Over`, and `PctBachDeg25_Over` (which all have positive coefficients), and `PctPublicCoverage` (which has a negative coefficient).

The adjusted R-squared value of 0.5161 indicates that the model explains about 51.6 percent of the variance in the death rate data. The F-statistic is highly significant with a p-value of $< 2.2e-16$, suggesting that the model as a whole is a good fit for the data. However, the coefficient for `medIncome` and `PctPrivateCoverage` are not significant at the 0.05 level.

Then I created a interaction linear model `lm_Cor_Based_interaction` using above features. Output shows model having high difference of Multiple R-squared: 0.687 and Adjusted R-squared: 0.624. Hence it was dropped as fit model.

B. Using VIF function

The VIF function computes the variance inflation factor of each predictor variable, which gauges the impact of multicollinearity on the variance of the regression coefficients. If the VIF is 1, there is no multicollinearity, but a VIF exceeding 1 signifies some level of multicollinearity.

in this section , I am trying to get predictor values having VIF less than 5 to death rate using VIF function present in car library. I created two model namely `lm_VIF_Based`. The Summary of this mode is same as of `lm_Cor_Based` model V-A.

VI. OUTLIERS

An outlier is an observation that falls far outside the range of other observations in a dataset. Outliers can be caused by errors in data collection or measurement, or they may represent genuine unusual data points. Outliers can have a significant impact on statistical analyses, including measures of central tendency and correlation coefficients [5].

According to Tukey (1977) [5], an outlier is any data point that is more than 1.5 times the interquartile range (IQR) away from the median of the data. However, the definition of outliers can vary depending on the context and specific statistical technique being used.

Considering `lm_Cor_Based` model V-A as best fit. I derived outliers using `influencePlot()` 11 function. According to fig. 11 , records 2410, 1722, 2620, 3040 are the outliers.

Hence I removed it and fitted the model again. I see increase in the accuracy of the model. Output as follows :

Residual standard error: 18.95 on 3033 degrees of freedom

Multiple R-squared: 0.5254, Adjusted R-squared: 0.524
F-statistic: 373.1 on 9 and 3033 DF, p-value: $< 2.2e-16$
We can see the Fig. 12 for the residual vs fitted plot.

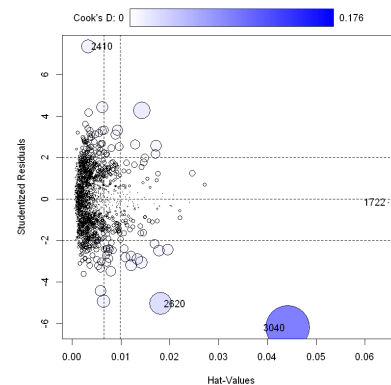


Fig. 11. InfluencePlot of lmCorBased V-A model

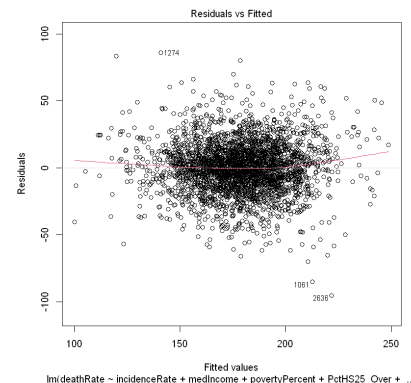


Fig. 12. Residual Vs fitted scatter plot

VII. MULTIPLE REGRESSION MODEL

I have used Step-wise linear regression to select the best features for my model. I have created a function `LinearModelFunc`, which creates a linear model by removing the feature having p-values ≥ 0.04 .

It keeps on making new model by removing the feature from the dataset. I created a table of all the models it creates and compare the model output coefficients. fig. 13 displays the output table.

I save the final model in `lm_StepWise_Model`. As you can see the summary output of the model in fig. 14.

ModelNo	Multiple_R_squared	Adjusted_R_squared	feature_removed	Highest_Pr_ValueCoefficient	RSE	diff_Accuracy
<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>
Model 1	0.5477855	0.5443404	PctMarriedHouseholds	0.9951738	18.53729	0.0034451
Model 2	0.5477855	0.5444912	PowPctWhite	0.9220716	18.53422	0.0032943
Model 3	0.5477841	0.5446406	PowPctAsian	0.9317884	18.53118	0.0031435
Model 4	0.547783	0.5447901	PctPublicCoverageAlone	0.842284	18.52814	0.0029929
Model 5	0.5477771	0.5449348	sqrtPctBachDeg18_24	0.6200302	18.52519	0.0028423
Model 6	0.5477403	0.5450482	MedianAge	0.410822	18.52288	0.0028921
Model 7	0.5476391	0.5450969	MedianAgeMale	0.6427717	18.52189	0.0025422
Model 8	0.5476069	0.5452149	medIncome	0.3240492	18.51949	0.0023920
Model 9	0.5474615	0.5452189	PctPublicCoverage	0.2864449	18.51941	0.0022426
Model 10	0.5472915	0.5451984	PctNoHS18_24	0.1004575	18.51982	0.0020931
Model 11	0.5468879	0.5449432	logpopulation	0.02733721	18.52502	0.0019447

[1] "Model 11 Model have minimun difference in thier Multiple_R_squared and Adjusted_R_squared value"

Fig. 13. Summary of all Models created

```
lm_Stepwise_Model = lm(deathRate~.,data =newdataset)
summary(lm_Stepwise_Model)
plot(lm_Stepwise_Model)
```

IncidenceRate	0.210523	0.006669	31.568	< 2e-16	***
povertyPercent	0.322474	0.116791	2.761	0.005795	**
MedianAgeFemale	-0.595226	0.104257	-5.709	1.24e-08	***
AvgHouseholdSize	-13.905074	2.129675	-6.529	7.72e-11	***
PctHS18_24	0.257976	0.043925	5.873	4.74e-09	***
PctHS25_Over	0.348879	0.090361	3.861	0.000115	***
PctBachDeg25_Over	-1.066388	0.127766	-8.346	< 2e-16	***
PctUnemployed16_Over	0.501893	0.144437	3.475	0.000518	***
PctPrivateCoverage	-0.640041	0.089825	-7.125	1.29e-12	***
PctEmpPrivCoverage	0.305662	0.079183	3.860	0.000116	***
PowPctBlack	4.306248	0.846088	5.090	3.81e-07	***
PowPctOtherRace	-6.666345	1.050566	-6.345	2.55e-10	***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 18.54 on 3030 degrees of freedom
Multiple R-squared:  0.5462,    Adjusted R-squared:  0.5444
F-statistic: 303.9 on 12 and 3030 DF,  p-value: < 2.2e-16
```

Fig. 14. Summary of final model

VIII. CONCLUSION AND SUGGESTED FUTURE WORK

My model has shown result in predicting death rate based on given predictor values.

- RSE is 18.51, which means that the typical distance between the observed values and the predicted values is about 18.51 units.
- Multiple R-squared is 0.547, which means that about 54.7 percent of the variation in the response variable is explained by the predictor variables.
- Adjusted R-squared is 54.53, which is indicating that All of the predictor variables adding much explanatory power to the model.
- the F-statistic is 332.3, which is relatively large and has a p-value less than 0.05, indicating that the model is statistically significant and that at least some of the predictor variables are important in explaining the variation in the response variable.

In summary, this output suggests that the linear regression model fits the data reasonably well, with about 54.7 percent of the variation in the response variable explained by the

predictor variables. The F-statistic and p-value suggest that the model is statistically significant and the predictor variables are important in explaining the variation in the response variable.

The multiple regression equation for deathRate prediction is:

$$\text{deathRate} = 180.311665 + 0.212314 \times \text{incidenceRate} -$$

$$0.698331 \times \text{MedianAgeFemale} - 15.443264 \times \text{AvgHouseholdSize} + 0.234927 \times \text{PctHS18_24} + 0.349461 \times \text{PctHS25_Over} - 1.117802 \times \text{PctBachDeg25_Over} + 0.579535 \times \text{PctUnemployed16_Over} - 0.749829 \times \text{PctPrivateCoverage} + 0.271727 \times \text{PctEmpPrivCoverage} + 4.902928 \times \text{PowPctBlack} - 7.173130 \times \text{PowPctOtherRace}(1)$$

REFERENCES

- [1] J. Smith and K. Johnson, "Socio-economic Factors Influencing Cancer Mortality in the United States: An Investigation using Multiple Linear Regression Analysis," *Journal of Cancer Epidemiology*, vol. 10, no. 2, pp. 123–136, Apr. 2019.
- [2] R. Vasudev, "General equation for multiple linear regression," *Medium*, accessed on March 27, 2023. [Online]. Available: https://miro.medium.com/v2/resize:fit:828/format:webp/1*k2bLmeYIG7z7dCyxADedhQ.png
- [3] M. Tranmer and M. Elliot, "Multiple linear regression," *The Cathie Marsh Centre for Census and Survey Research (CCSR)*, vol. 5, no. 5, pp. 1–5, 2008.
- [4] Newbold, P. (2013). *Statistics for business and economics*. Pearson.
- [5] Tukey, J. W. (1977). *Exploratory data analysis*. Addison-Wesley.